

# Building Open and Reproducible Geospatial Simulations with Python

## DisSModel — Discrete Spatial Modeling Framework

Sérgio Souza Costa · LambdaGeo / UFMA

Cellular Automata

System Dynamics

LUCC · Mangroves

LUCC · CLUE

## AGENDA

# What we will cover today

**01** Contextualization

3 min

**02** The Problem

6 min

**03** DisSModel: A Discrete Spatial  
Modeling Framework for Python

5 min

**04** The DisSModel Architecture

5 min

**05** Laboratory Models

3 min

**06** Empirical Models

4 min

**07** What Do We Gain with DisSModel?

2 min

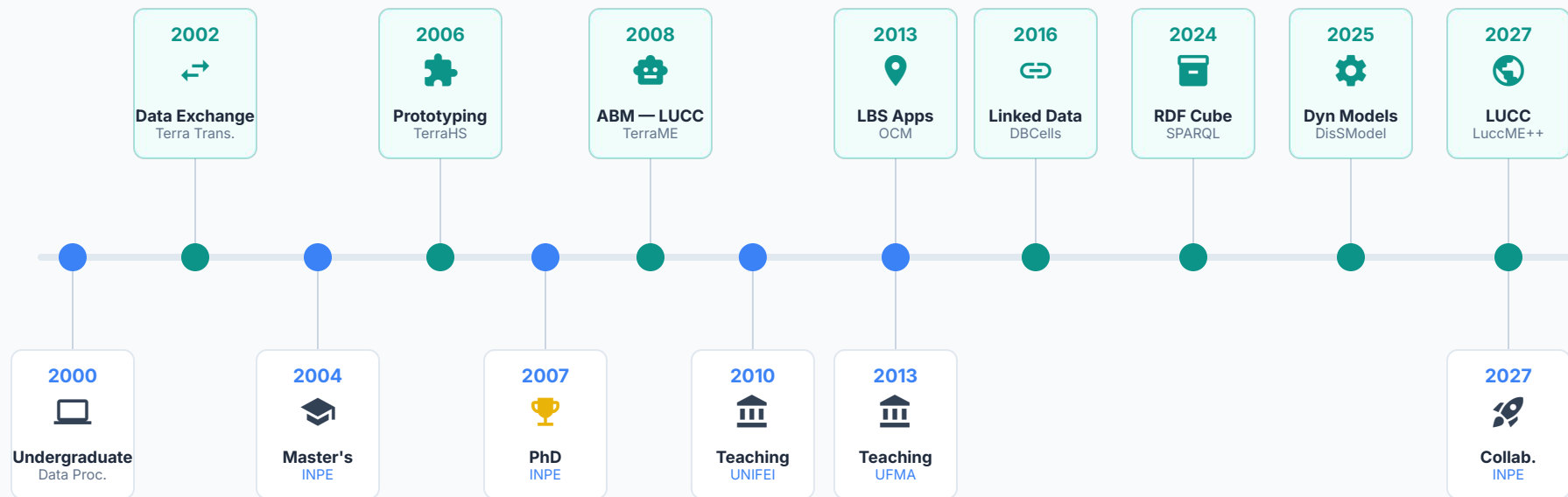
**08** Next Steps

2 min

# Career and Research Evolution

● Research Projects

● Academic & Professional



Focus on Spatial Modeling and Geographic Data (Open Science)

# Geospatial Simulations

## Land Use and Cover Change (LUCC)

LUCC models how and why the land surface changes over time. It is the mathematical language of human consequences on the territory.

### Deforestation

Conversion of native forest to agricultural use.

### Coastal Dynamics

Mangroves vs. urbanization and sea level rise.

### Urban Expansion

Growth of cities over rural and natural areas.

### Agricultural Frontier

Agribusiness advance over the Cerrado and the Amazon.

#### FOCUS: PROCESS-BASED GEOSPATIAL MODELING

### Expectation Alignment

This presentation focuses exclusively on **Geospatial Models** instead of general-purpose Machine Learning models.

# Key Brazilian Geospatial Tools

## Dinamica EGO

UFMG · since 2002

Visual GUI platform for spatially explicit LUCC modeling.

### STRENGTHS

- No-code visual interface
- Academic benchmark

### LIMITATIONS

- Manual reproducibility
- Scalability issues

## TerraME

INPE · since 2005

Lua simulation framework for cellular automata models.

### STRENGTHS

- Native Agent/Space logic
- PostGIS integration

### LIMITATIONS

- TerraLib is no longer supported
- TerraME last release (2020)
- No BDC/STAC support

## LuccME

INPE · built on TerraME

Modular framework based on Demand/Allocation architecture.

### STRENGTHS

- Interchangeable parts
- CLUE methodology

### LIMITATIONS

- Inherits Lua constraints
- Weak Python link

# The reproducibility crisis in computational science

Baker (2016), Nature · NAS (2019) · Stodden et al. (2016), Science · Kracczyk et al. (2021)

## Baker, Nature 2016

70%

of researchers could not reproduce experiments from other groups.

## NAS, 2019

National Academies, commissioned by the U.S. Congress. Formal report defining computational reproducibility as a pillar of modern science and establishing institutional recommendations.

### Code doesn't run

Undocumented dependencies, incompatible versions, **implicit environments** — the experiment only works on the author's machine.

### Inaccessible data

Input data not versioned or not published. **Without original data**, the result cannot be audited or replicated.

### Opaque parameters

Critical model configurations are in ad hoc scripts or in the **researcher's memory**, not in formal auditable records.

# Challenges in Dynamic Geospatial Modeling

Spatial simulations have additional challenges beyond general computational reproducibility

01

## Geospatial data is large and volatile

Rasters and vectors with specific CRS, varying resolutions and version history. A different GeoTIFF silently masks calibration errors.

02

## Models couple science and infrastructure

The CA or LUCC logic gets mixed with file I/O, visualization and loop control — impossible to reuse without rewriting.

03

## Experiments are not traceable

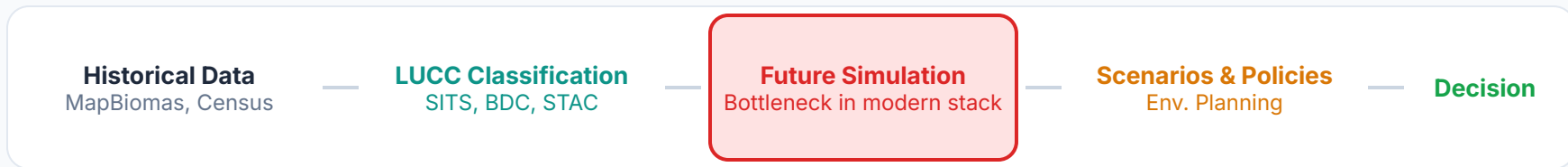
Which model version? Which parameters? Which input dataset? Without formal records, the result is irreproducible by design.

04

## Scale breaks the scientific workflow

The researcher calibrates in the notebook, but running scenarios at scale requires rewriting the code — or abandoning production execution.

# Why doesn't LambdaGEO propose modernizing TerraME?



|                   | TerraME / LuccME                      | DisSModel (The Proposal)          |
|-------------------|---------------------------------------|-----------------------------------|
| Language          | Lua + TerraLib (no longer maintained) | Pure Python (Geopandas, NumPy)    |
| Reproducibility   | No formal record                      | Git + SHA-256 per execution       |
| Cloud Integration | Isolated local execution              | Cloud Native (API, Minio, Docker) |

*DisSModel preserves the TerraME modeling philosophy in a modern ecosystem.*



# The Researcher's Journey: End-to-End Workflow

Modules of the @dissmodel Organization

## PHASE 1 DEVELOPMENT & PACKAGING

### 1. Creation

Model design using the base library (model, environments, geo models, rules).

### 2. Packaging

Logic is fully encapsulated in *executors* (validate, load, execute, save).

### 3. Tests

The model is validated locally in an incredibly agile way via CLI tool.

### 4. Publishing

Library source code made available in independent GitHub repositories.

## PHASE 2 INTEGRATION & EXECUTION

### 5. Integration

Researcher creates a .toml manifest and submits a PR to the catalog.

### 6. Curation

Maintainers review compliance and security before official incorporation.

### 7. API Execution

Platform manages dependencies and exposes results via dynamic API.

*"Science should not need to be rewritten to go into production."* — DisSModel Principle

# Core Pillars of DisSModel

Essential foundations for robust and scalable scientific modeling.

## Reproducible Research

Sharing data and code for external verification.

### IMPLEMENTATION:

- Executor + TOML: separation of code/params
- Docker: isolated execution
- SHA-256: auditable records

*Claerbout (1992) · Stodden (2016)*

## FAIR Software

Findable, Accessible, Interoperable, and Reusable.

### KEY PILLARS:

- F:** Zenodo DOI + reprod. package
- A:** REST API + MinIO/S3
- I:** GeoPandas + Xarray + GeoTIFF
- R:** Versioned TOML manifest

*Wilkinson et al., 2016*

## Software Engineering

Bridging Scientists and Production Engineers.

### LAYER SEPARATION:

- **Science:** logic remains pure
- **Platform:** APIs and infrastructure
- **SimOps:** Scientific Operations

*Baxter et al. (2012) · Wilson (2017)*

"Science should not need to be rewritten to go into production." — **DisSModel Principle**

# Main Repositories

Modules of the @dissmodel Organization



## Base

**3 repositories**

Provides the model's core engine, platform infrastructure, and official catalog.



## Lab Models

**2 repositories**

Focused on cellular automata and dynamic systems for testing, teaching, and learning.



## Empirical Models

**2 repositories**

High-fidelity models previously implemented in TerraME, now available as reproducible Python workflows.



## Interfaces

**1 repository**

A QGIS plugin that runs the BR-MANGUE model natively — no API calls, no middleware, simulation inside the GIS environment.

# Base Components

## **dissmodel**

### LIBRARY (Core Framework)

Central simulation engine for local scientific model development and execution.

[github.com/DisSModel/dissmodel](https://github.com/DisSModel/dissmodel)

```
pip install dissmodel
```

## **dissmodel-platform**

### PLATFORM (Cloud-Native)

Open and scalable infrastructure for distributed cloud experiment execution.

[github.com/DisSModel/...-platform](https://github.com/DisSModel/...-platform)

```
docker compose up --build
```

## **dissmodel-configs**

### MODEL CATALOG

The single source of truth. Centralizes configuration files (manifests in .toml).

[github.com/DisSModel/...-configs](https://github.com/DisSModel/...-configs)

```
pull requests
```

*"Science should not need to be rewritten to go into production" — **dissmodel** runs the same code locally and on the remote **platform**.*

# Core Framework

## core

### Simulation Engine

Discrete clock +  
execution cycle via  
Environment & Model.

*Environment, Event  
dispatch, step control.*

## geo

### Spatial Substrate

Dual design: Vector  
(GeoDataFrame) +  
Raster (NumPy).

*Queen/Rook and  
Moore/Von Neumann  
neighborhoods.*

## executor

### Execution Layer

ModelExecutor ABC +  
SimulationRegistry.

*Local CLI and Remote  
Workers — same code.*

## io

### Storage I/O

Local file and MinIO/S3  
read/write operations.

*Checksums and  
provenance.*

## vis

### Visual Observers

Chart, Map,  
RasterMap,  
display\_inputs.

*Decoupled Observer  
pattern.*

"Science should not need to be rewritten to go into production" — **DisSModel Principle**

# DisSmodel-platform - Server Side

## FastAPI Gateway

### API-Driven Execution

- Programmatic REST calls
- Legacy dashboard integration
- Validation via Pydantic

## Docker Workers

### Subprocess Isolation

- Jobs in isolated containers
- No dependency conflicts
- Redis Queue scalability

## Jupyter Lab

### Experiment Service

- Interactive experimentation
- Direct data visualization
- Rapid model prototyping

## MinIO Storage

### Agnostic Storage

- S3-compatible integration
- Unified spatial writing
- Zarr and GeoTIFF support

*The infrastructure orchestrates **FastAPI, Redis, and Docker** to ensure science does not need to be rewritten for production.*

# DisSModel-configs - The Central Catalog:

## Single Source of Truth

To keep the ecosystem scalable and secure, we separate the model code from its publication.

The catalog stores only declarative manifests, ensuring full traceability and governance.



### Manifestos .toml

Each model has a centralized configuration file.



### Versioning

Points to the specific *\*release\** or *\*tag\** on GitHub.



### Parameterization

Defines default limits consumed by the API.

```
configs/models/coastal_raster.toml

[model]
executor_module = "br-mangue.executor"
name = "br-mangue"
package = "git+https://github.com ..."

[model.parameters]
end_time = 88
taxa_elevacao = 0.5
altura_mare = 6.0
acrecao_ativa = false
resolution = 100.0
crs = "EPSG:31984"
```

# Practical Examples & Implementations

Now that the base repositories and server-side architecture are established, let's explore the practical laboratory models built upon this foundation.



## Spatial Models

### Complex systems using Cellular Automata (CA)

- Localized spatial interactions
- Game of Life & Forest Fire
- Built on GeoDataFrames



## Temporal Dynamics

### System Dynamics (SysDyn) Framework

- Flows, stocks, and feedback loops
- SIR & Predator–Prey models
- Interactive Streamlit explorers

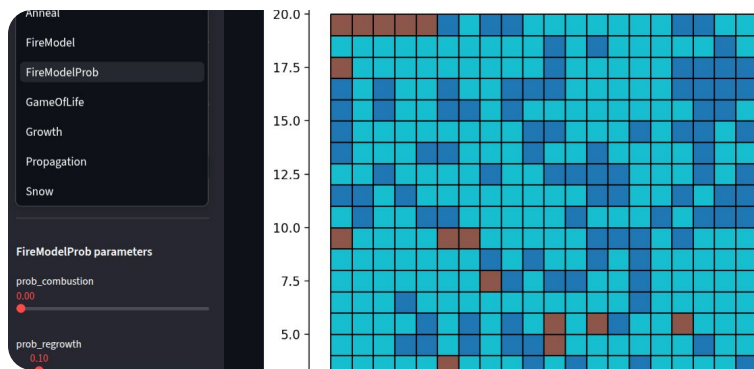


# Cellular Automaton: Game of Life

## Model Definition

```
class GameOfLife(CellularAutomaton):  
    def setup(self) → None:  
        self.create_neighborhood(strategy=Queen)  
  
    def rule(self, idx) → int:  
        state = self.gdf.loc[idx, 'state']  
        n = self.neighbor_values(idx).sum()  
        if state == 1:  
            return 1 if 2 ≤ n ≤ 3 else 0  
        return 1 if n == 3 else 0
```

## Model Execution & Results



```
gdf = vector_grid((20, 20), attrs={"state": 0})  
env = Environment()  
GameOfLife(gdf).initialize()  
Map(gdf, column="state")  
env.run(steps=10)
```

# System Dynamics: SIR Model

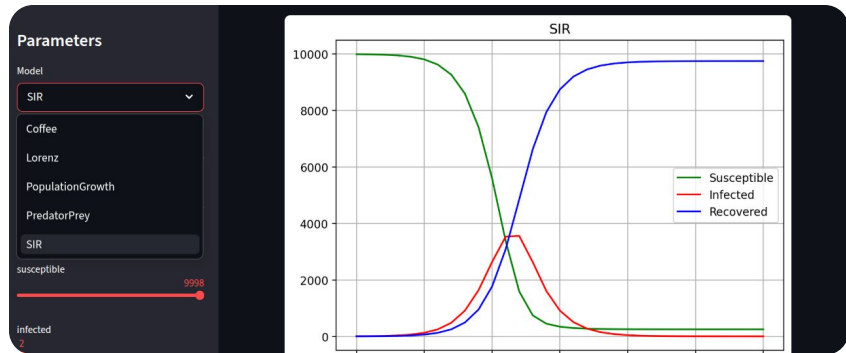
## Model Definition



```
class SIR:
    def execute(self):
        total = S + I + R
        alpha = contacts * probab
        new_inf = I * alpha * (S/total)
        new_rec = I / duration

        S -= new_inf
        I += new_inf - new_rec
        R += new_rec
```

## Results & Implementation



```
env = Environment()
SIR(9998, 2, 0, 2, 6, 0.25)
Chart(show_legend=True, show_grid=True, title="SIR Model")
env.run(30)
```

# Real-World Empirical Models

## From Theory to Application

Transitioning from laboratory simulations to complex, empirical scenarios. Explore how DisSModel addresses coastal dynamics and land use change in real-world environments.



### Coastal Management

Modeling mangrove migration and flooding risks. Analysis of sea-level rise scenarios in the Maranhão Coast.



### Land Cover Change

Simulating agricultural expansion and urban growth. Empirical validation with real-world land use datasets.

# Coastal Case Study (brmangue)



## Study Area

Maranhão Coast, Brazil. Vector spatial grid.



## Scientific Basis

Bezerra et al. (2014).



## FloodModel

Flooding due to sea level rise scenarios.



## MangroveModel

Migration by tide and soil type dynamics.



```
env = Environment(start_time=2008, end_time=2030)
# Add specialized models
FloodModel(gdf=gdf, elevation_rate=0.5,
use_attr='usage')
MangroveModel(gdf=gdf, tide_height=6.0,
soil_attr='soil')

env.run()
```

## Performance Benchmark

SUBSTRATE | MS/STEP | SPEEDUP

Vector: **1241.7 ms** (1.0x)

**Raster: 31.5 ms** (39.4x ⚡)

Scale: 97,309 cells

Accuracy vs TerraME: **99.8% Match**

# Coastal Case Study (brmangue)

## Plugin Feature

## QGIS Integration

Dedicated plugin for the brmangue model, enabling streamlined simulation and visualization of coastal dynamics within the QGIS environment.

- Intuitive interface for parameter input
- Direct visualization of model outputs
- Seamless integration with vector grids

The screenshot shows the QGIS interface with the BR-MANGUE Coastal Simulation Platform plugin installed. The main map displays a coastal area with a yellow and green simulation output. The plugin window is open, showing configuration options for the DisModel Platform, dataset input, model parameters, and visualization bands. A 'Painel de Experimento' window is also visible, displaying the FAIR citation, provenience and hashes, resolved specifications, and execution logs.

**BR-MANGUE — Coastal Simulation Platform**

**Servidor DisModel Platform**

URL:

API Key:

**Dataset de Entrada**

URI do dataset:

Formato:

**Parâmetros do Modelo**

End time (steps):

Taxa de elevação:

Altura da maré:

Acreção ativa: ☐

**Bandas a Visualizar**

☒ uso (uso e cobertura) ☐ solo (tipo de solo) ☐ alt (elevação)

**Status**

Aguardando submissão.

Carregando pré-visualização de s3://dissmodel-inputs/ilha\_maranhao\_epsg31983.tif...

Erro ao carregar pré-visualização: O servidor não retornou uma URL válida para download.

Carregando pré-visualização de s3://dissmodel-inputs/ilha\_maranhao\_epsg31983.tif...

**Painel de Experimento — 41f50dd0**

**Citação FAIR**

DisModel v1.0 (spec: brmangue@2ef1bbc, output sha256: 964ce4ff1d5c...)

**Proveniência e Hashes**

Status: COMPLETED

Model Commit: 2ef1bbc

Input SHA256: 1b06982d2dc04c4d8211dc717afa89f76d932e575fbafe90e6af44e774e30c

Output SHA256: 964ce4ff1d5c81d1a625740f4cf1ebb26d00d307f18ab92f2f2bcb235e5dc94e

**Resolved Spec (Parâmetros)**

**Logs de Execução**

```
Record created - model=brmangue commit=2ef1bbc
Dispatching to subprocess...
Loaded GeoTIFF: shape=(418, 405) start=1 crs=EPSG:31983
Running steps 1 - 88...
Simulation complete
Saved to s3://dissmodel-outputs/experiments/41f50dd0b304a63a9e931c93a2d53e6/output.tif
Saved profiling artifact - s3://dissmodel-outputs/experiments/41f50dd0b304a63a9e931c93a2d53e6/profiling_41f50dd0.nd
Saved record JSON - s3://dissmodel-outputs/experiments/41f50dd0b304a63a9e931c93a2d53e6/41f50dd0.record.json
Completed - val=0.00s | loaded=15s | run=0.03s | save=0.04s | total=7.12s
```

# DisSLUCC-Continuous: Land Use & Cover

## Continuous Modeling

The package implements the CLUE-type allocation algorithm (Veldkamp & Fresco, 1996). Cells receive *fractional proportions* instead of discrete classes.

## Evaluation Metric

For continuous outputs, the **MAE (Mean Absolute Error)** was adopted to directly evaluate modelling error (Willmott, 2005).

## 1. Demand

Magnitude of change per timestep.  
Pre-computed via historical CSV.

```
DemandPreComputedValues(  
  annual_demand=load_csv( ... )  
)
```

## 2. Potential

Change suitability via linear  
regression with driver variables.

```
PotentialLinearRegression(  
  gdf=gdf, drivers=[ ... ]  
)
```

## 3. Allocation

CLUE-like distribution based on  
potential and constraints.

```
AllocationClueLike(  
  demand, potential  
)
```

## Experimental Setup

DATASET  
**LuccME AC (2008-2014)**

TIME STEPS  
**6 Steps**

RESOLUTION  
**6,574 cells**

# DisSLUCC - Validation and Performance Results

## Performance (6 steps · 6,574 cells)

**3.9×**

**SPEEDUP RASTER VS VECTOR**  
(31.6 ms vs 122.5 ms)

## Accuracy vs TerraME

**0.002276**

**MAE VS TERRAME · BOTH SUBSTRATES**  
Vector = Raster · MAE 0.002276

**Technical Highlight:** Raster substrate achieves 3.9× speedup over Vector (31.6 ms vs 122.5 ms/step). Both substrates produce identical output (MAE = 0.000) compared to each other, confirming the raster optimisation preserves full algorithmic correctness.

Dataset: LuccME Package - AC - 2008–2014 · 6 steps · 6,574 cells · tolerance 0.01

# TRANSFORMING GEOSPATIAL SCIENCE



## Hidden Engineering

The complex cycle of data orchestration runs seamlessly.



## Full Traceability

Guaranteed data provenance for every simulation stage.



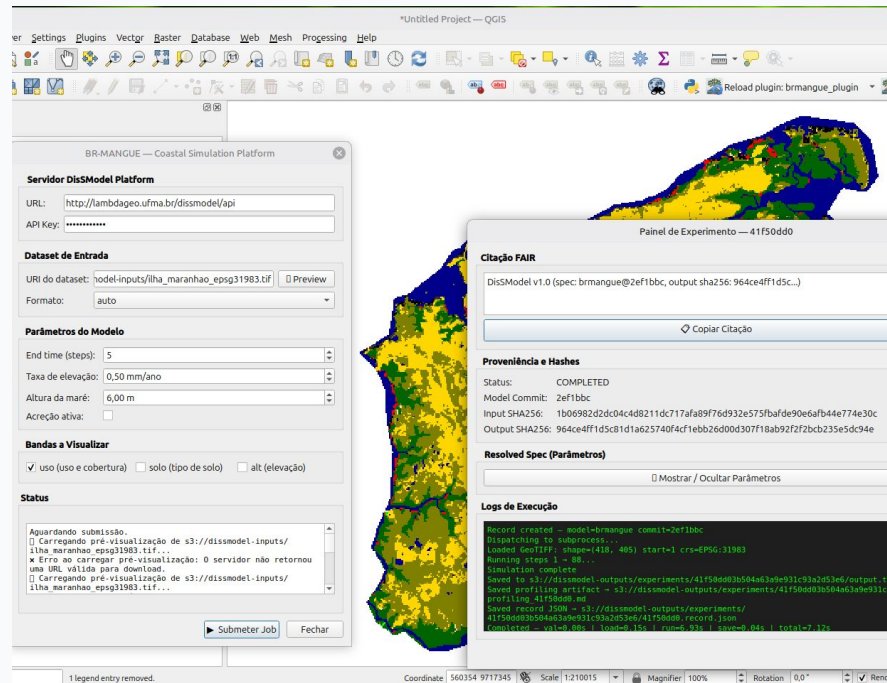
## Direct Delivery

Dynamic results delivered directly within GIS tools.



## End-to-End Focus

Researchers focus on science, users on decisions.





# DisSModel Roadmap 2026–2028

2026 S1

2026 S2

2027 S1

2027 S2

2028

## JOSS Publication

Repo organisation &  
PyPI v0.5  
Formalisation at UFMA  
Research plan  
submissions

## Project Kickoff

CLUE & CLUE-S  
(PIBIC)  
DisSLUCC (PIBIC)  
BRMangue JOSS  
DisSModel v1.0

## BDC Integration

STAC source reader  
Xarray (time, y, x)  
ExpRecord → MinIO

## Dask / Ensemble

Lazy eval (Pangeo)  
Parallel scenarios  
Native Zarr output

## Book + Community

Geospatial DS book  
PT-BR Edition  
Workshop INPE/SBSR

\* QGIS plugins (*qgisparql-triple2layer* / *layer2triple*): joint JOSS submission under evaluation.

## Strategic Partnership: DisSModel & LuccME++

**2027/28** — DisSModel will serve as the innovation laboratory for the evolution of **LuccME** at CCST/INPE.



**DisSModel**

Partnership &  
Evolution  
→



**Future of LUCC**

*TerraME was the starting point* — **DisSModel** advances as the foundation for **LuccME++**

# Thank you!

*DisSModel — open source, open science*

**GitHub**

`disssmodel.github.io`

**Docs**

`disssmodel.github.io/disssmodel/`

**PyPI**

`pip install disssmodel`

**Contact**

`sergio.costa@ufma.br`

*"Science should not need to be rewritten to go into production."*  
— DisSModel Principle

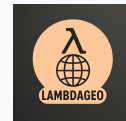
`disssmodel`

`disssmodel-ca`

`disssmodel-sysdyn`

`brmangue-disssmodel`

`DisSLUCC-Cont.`



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